

OAL

Temporal Claim Provenance and Drift Laboratory (TCPDL)

Independent research framework, reference model, and first test run

Version record

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Preparer	Claude (Anthropic). Independent contribution. Not a merge, average, or consensus of other contributors.
Topic origin	Nathan's non-factual thought experiment on time: the unit of time in a virtual environment; validating an event that uses time as a reference marker; tracking and reconciling change that is not uniform across territories; the 100-year token; the one-hour block; light-years across time zones; and an event at the origin of light. Purpose: to understand time's characteristics and mechanisms well enough to support the Time Wars worldbuilding. It is not a claim about any real or virtual system.
Documents reviewed	Six-contributor packet (ChatGPT, Claude, Gemini, Perplexity, Copilot, Grok). Then two independent frameworks: Perplexity, "Independent Temporal Integrity Research and Testing Framework" (PDF); Grok, "OAL Temporal Lab Framework" (2026-09-29 DRAFT, MD and DOCX).
Purpose of this document	To review those two frameworks; to define a third, independent laboratory; and to run a first deterministic test campaign on a small closed fictional world so that the framework has measurements behind it, not only specifications.
Hypotheses tested	H1–H7 (Section 6). All are model-internal: they test the framework inside the reference model, not any real system.
Relationship to my earlier response	That compact seven-section response is superseded here. Its three-stamp validation model was incomplete, as Grok's framework correctly noted (Section 3.2).
Companion file	mw1_reference_model.py (Reference Model MW-1 v0.1). Deterministic. Re-running it reproduces every table in Section 8.
Source of intent	Nathan's original dictation remains the governing statement of intent. Section 2 restates it for traceability only and does not replace it.

1. Reader's guide

Every proposition carries one of four labels. The first three are the scheme Perplexity's framework introduced, which I adopt because it keeps established concepts separate from invented ones. The fourth is mine.

Label	Meaning	Example
E	Established analogue: accepted practice in physics, standards, or computing.	A light-year is a distance, not a duration.
D	Design rule: chosen for this project because it improves consistency or auditability.	Corrections are appended, never overwritten.
S	Speculative extension: a fictional capability whose behavior must be stipulated.	A token permits traversal into a closed block of time.
M	Model result: computed in the reference model MW-1. It holds only inside that toy model.	A strict reader made 0 wrong answers over 308 questions.

I do not fact-check the fictional mechanisms. Where I test something, I test whether the framework behaves coherently under its own declared rules.

2. What was asked

The question underneath the original dictation was never "what is time?" It was: **what information must accompany a time reference before it can be trusted as a claim, and how do you keep that claim usable when clocks, territories, observers, and rulesets do not change together?**

The dictation then poses concrete probes of that question. Each is treated in this document:

- the unit of time in a virtual environment, and how events that use time as a marker are validated, tracked, and reconciled (Sections 5, 8);
- non-uniform time change across territories (Sections 7, 8.4);
- a token that buys every time change in 100 years (Section 8.6);
- a one-hour block of time (Sections 7, 8.1, 8.5);
- light-years across time zones, and an event at the origin of light (Section 8.6);
- implications for the Time Wars (Section 10).

3. Review of the two independent frameworks

3.1 Perplexity: Temporal Integrity Test Framework (TITF)

What it is. A test-methodology program built on the packet's convergence. It proposes a versioned temporal claim graph, 15 invariants, a 9-gate validation protocol, a 10-step reconciliation protocol, a 10-dimension test matrix with pairwise coverage, 8 test oracles, 5 suites (one-hour block, century ledger, light and observation, jurisdiction, adversarial), 12 metrics, and a 7-phase plan.

Strengths.

- It states what cannot be established: no proof of a "true" time, and no complete century audit without a bounded corpus of territories and evidence.
- It separates evidentiary findings from authority decisions.

- It uses several oracles instead of one aggregate score, and makes safety-critical invariants pass/fail gates.
- Corpus-bounded completeness is the right discipline for the 100-year token.

Gaps.

- **No run.** It says the material is "not yet ready for a single full test" and defers to Phases 0–2. That is honest, but it means none of its 15 invariants has been exercised.
- **Invariant 3 needs a scope.** "Accepted mappings cannot place B before A" is true of order coordinates, but false of display labels: a backward clock change legitimately gives a later event an earlier label (MW-1 test A3).
- **Sufficiency is per question, not per claim.** The nine gates validate each claim. They do not say which fields a claim needs for which question, so they cannot say when a partial record is still good enough.
- **No traceability.** It does not say which invariant a given test protects, so overlap among 15 invariants, 9 gates, and 8 oracles cannot be checked.
- **Nothing on hand-off loss.** Its Phase 5 compares contributor outputs, but no instrument measures what a claim loses as it is summarized, re-serialized, or passed between contributors.

3.2 Grok: OAL Temporal Lab Framework (TLF)

What it is. A lab charter with a forbidden assumption ("what time is it, really, everywhere?"), ten primitive objects, four unit kinds, eight integrity principles, an event-record schema, a reconciliation procedure, three protocols (century token, one-hour block, light-years), a nine-row operations table, a 12-item attack catalog, a ten-test battery (T1–T10), a ten-item architect decision register, and a "verdict on the team packet".

Strengths.

- The charter discipline. The lab may not assume it knows the current time everywhere.
- Never using one verb for relabel, remap, reclock, reobserve, resettle, rewind, branch, merge, and rewrite. I adopt this.
- "Unknown is data", and treating a successful fraud or a lost archive as a valid lab outcome rather than an incomplete test.
- Its critique of my earlier response is fair. A happened/observed/recorded model omits ruleset version, frame, uncertainty, and the dependency cone.

Gaps.

- **Expectations without measurements.** T1–T10 say what a system "must" do, but no system, ground truth, or run is supplied.
- **Unit kind is typed on the marker.** One civil label can serve a presentation question and, with the rules attached, an ordering question. The kind belongs to the question being asked, not to the marker.
- **Fail-closed on underspecified input** (T6 and T7 "refuse"; T1 "reject or mark incomplete"). That is right for granting rights. It is wrong for reading records, where a partial answer with an explicit completeness flag is more useful (H7).
- **The verdict section compresses.** It reduces each contributor to an adopt/reject line. On the points I checked against the packet it is accurate, but it is still a compression; the original sections remain the source.

3.3 Where the two frameworks and this one stand

Topic	Perplexity TITF	Grok TLF	TCPDL position
Underspecified input	Annotate: surface the ambiguity (invariant 6).	Mostly refuse (T6, T7); T1 allows "mark incomplete".	Grants fail closed. Reads annotate with a completeness flag. (H7)
Unit typing	Not typed.	Four unit kinds on the marker.	The kind attaches to the question, not the marker. (5.3)
Order invariant	Causal monotonicity of mappings.	Happened-before via the causal graph.	Scope the invariant to order coordinates. Label order is presentation. (A3)
Authority vs evidence	Record decisions separately from findings.	Settle per domain.	Three ledgers. A settlement can cite claims but cannot write to them. (5.2)
Execution	Phases 0–2 first, then a campaign.	A closed-world run is described (§15) but not performed.	Run a micro-world now; use failures to see which canon decisions matter. (8)
Test design	Oracles, pairwise matrix, property-based tests, fault injection.	Fixture battery and attack catalog.	Ablation traceability, hand-off drift chain, and commutation checks.
Master clock	Optional; declared mappings.	None until a settlement act.	None in-world. The test harness carries a hidden reference R, declared as such. (7)

4. Laboratory charter

Name. Temporal Claim Provenance and Drift Laboratory (TCPDL).

Thesis. In this lab, time is not the specimen. The **temporal claim** is the specimen. The lab measures what a claim can still support after it is relabeled, remapped, corrected, summarized, handed to another reader, or settled by an authority, and whether the reader can tell when it no longer supports the answer being demanded of it.

Question the lab may ask. For a given claim and a given question, is the claim sufficient to answer it? If not, what is missing, and what does a reader do with the gap?

Question the lab may not assume is answered. What time it really is, everywhere. What the "true" ordering of distant events is. Whether any single authority's reconciliation is measurement rather than settlement.

Standing constraints (carried from OAL lab language where relevant):

- Do not assume all entities share one clock apparatus, one tick, or one calendar.
- Do not assume every discrepancy is detectable, discernible, or addressable. A successful forgery and a lost archive are valid outcomes.
- Presentation state, environmental state, and current process state may disagree. Record all three; do not collapse them.
- Drafts are not canon. Contributions are kept separate and provenance-tagged, not averaged.

Non-goals. Proving one correct definition of time. Certifying a complete real-world history of clock changes. Any claim that history, light, or an existing platform can be purchased or rewritten.

5. Concepts and architecture

5.1 Event, claim, interpretation, settlement

Object	Definition	Mutable?
Event	The asserted occurrence. It has an identity; it is not a timestamp. (D)	The fictional event itself is a world-rule question (decision 9). The record of it is not.
Claim	An assertion that an event occurred, with the markers, clock, ruleset, branch, observers, and evidence that accompany the assertion. (D)	Never. A correction is a new claim linked to the old one.
Interpretation	A mapping or reading applied to claims: a ruleset version, a conversion function, a correction, a reclassification. (D)	Append-only. A new interpretation does not replace an old one.
Settlement	An authority's decision about which reading governs a dispute. (D)	Append-only, and cannot modify claims or interpretations.

5.2 Three ledgers and five rules

Perplexity separates evidentiary findings from authority decisions; Grok separates operations. I make both structural, with three append-only ledgers:

- **Claim ledger (CL):** assertions, observations, and corrections as new rows.
 - **Interpretation ledger (IL):** rulesets, mappings, remaps, and reclassifications, each referencing CL rows.
 - **Settlement ledger (SL):** authority decisions, each referencing CL and IL rows.
1. **LR1.** A CL row is never modified or deleted. A correction is a new row that references the old one. (D)
 2. **LR2.** An IL row names the CL rows it interprets, the mapping version, the author, and the evidence. Changing a mapping never rewrites a CL row. (D)
 3. **LR3.** An SL row may cite CL and IL rows but can write to neither. A settlement presented as a measurement is a laundering failure (DR5, below). (D)
 4. **LR4.** Every answer lists the rows it used. (D)
 5. **LR5.** Every answer is three-valued: an answer, INDETERMINATE (evidence insufficient for this question), or INVALID (the claim is malformed, e.g. a label inside a skipped interval). Each carries a residual list. (D)

Every Time War verb then maps to exactly one ledger write (Section 10.2).

5.3 Question classes and interpretation sufficiency

A claim is not "complete" in the abstract. It is **sufficient for a question** if every reading consistent with the fields it retains gives the same answer. I compute this with a candidate-set rule: list every reference instant a claim could denote, then ask whether all candidates agree on the question. (D, M)

Question class	Minimum fields	If the fields are missing
INSTANT: when, on a declared reference?	occurrence reference, or (label + clock + ruleset version)	Folds give two candidates; gaps give none; missing ruleset gives unknown.

Question class	Minimum fields	If the fields are missing
ORDER-REF: which came first on the reference?	candidate sets for both claims, or a causal edge	Overlapping candidate sets cannot be ordered.
ORDER-CAUSAL: could A have influenced B?	branch, declared causal edges	Time order is silently substituted for causality.
MEMBERSHIP: is it inside interval I?	interval definition (frame, closure) + candidate set	Ambiguous instants may still agree: sufficient here, insufficient for INSTANT.
DISPLAY: what did the local clock read?	clock, ruleset version, instant	The current ruleset is assumed instead of the one in force.
OBSERVED: when did an observer learn it?	observation record, channel, delay	Receipt time is mistaken for occurrence time.
AS-OF: what did the system believe at K?	CL rows with record times, correction links	Overwrite makes past belief unrecoverable.
RIGHT: did it satisfy a rule or treaty?	ruleset version, and which dispute rule governs	One verdict is returned and its rule-dependence is hidden.

The consequence: the minimum viable record is a matrix of fields by question class, not one schema. This resolves the refuse-or-annotate divergence between the other two frameworks. A read is answered for every question the record is sufficient for, and flagged for the rest.

5.4 Drift

Drift is a change in what a claim supports after a transformation: relabeling, remapping, re-serializing, summarizing, merging, or handing to another reader. Measured concretely: for a fixed set of questions, the count of answers that change from determinate to INDETERMINATE (honest loss), or from correct to wrong (dangerous loss).

Code	Drift type	Signature
DR1	Field loss	Fields required by a question class are dropped.
DR2	Default substitution	A missing field is filled with an assumption: current ruleset, one universal clock, first candidate.
DR3	Qualifier stripping	A scoped statement loses its scope: a timeline, a frame, or a limiting clause.
DR4	Unit-kind conflation	An order count is treated as a duration, or a label as an instant.
DR5	Authority laundering	A settlement is presented as evidence or measurement.
DR6	Provenance flattening	Several sources are merged into one consensus statement; who said what is lost.
DR7	Precision inflation	An output is more precise or confident than its inputs.
DR8	Identity merge or split	Two events collapse to one record, or one event splits into two.
DR9	Silent decision	An open architect decision is answered implicitly by a default.

5.5 Two readers

The lab uses two reference readers so that drift can be measured. The **strict reader** applies the candidate-set rule and returns INDETERMINATE whenever candidates disagree or the fields are missing. The **default-assuming reader** fills every gap: it reads a label as a reference time, assumes the latest ruleset, and takes the first candidate on a fold. Neither is a claim about any real system. Together they turn "this record is underspecified" into a number.

6. Hypotheses

All hypotheses are model-internal (label M when tested) and each states what would falsify it.

ID	Hypothesis	Falsified if
H1	Sufficiency is question-relative: some claim is insufficient for one question class and sufficient for another.	Every claim is sufficient or insufficient for all question classes together.
H2	A strict reader never manufactures certainty; a default-assuming reader does.	The strict reader returns any wrong answer, or the default-assuming reader returns none.
H3	An append-only ledger preserves reconstruction of past belief, remaps, and rule-dependent verdicts. Overwrite destroys them.	Removing append-only behavior breaks none of the tests.
H4	Whether two changes commute depends on the pair of operations and the coordinate they are scheduled in, not on operation type alone.	The same two operation types always commute, or never commute, in every scheduling coordinate.
H5	The same evidence yields different verdicts under different dispute rules; a single-verdict ledger hides this.	All dispute rules return the same verdict.
H6	A dependency-cone propagates to strictly fewer records than a universal recompute.	The cone equals the whole record set.
H7	Fail-closed grants and annotate-and-flag reads yield no rights expansion and maximum usable information.	Removing the authority model does not change either outcome.

7. Instrument: Reference Model MW-1 v0.1

MW-1 is a closed, deterministic, fictional micro-world of three territories, one observer station, and eleven events. It is measured in fictional **reference minutes (R)**. It makes no claim about any real clock or system. It exists to give the framework ground truth to be wrong against.

Declared asymmetry. MW-1's test harness holds a hidden reference R for every event, used only to score answers. No in-world reader is given R unless a claim carries it. In a canonical universe, an equivalent ground truth may not exist, in which case INDETERMINATE answers can never be checked (Section 12).

Territories and clocks

Territory	Clock behavior (display vs reference R)	Purpose
Harbor	Shows 09:00 at R0. Rate ×1 until R30, then ×2. Ruleset v3. A decree announced at R90 (v4) reinterprets all displays +60 min retroactively.	Rate change and retroactive reinterpretation
Ridge	Display = R until R90; then jumps back 60 min (fold : labels 00:30–01:30 occur twice); at R200 jumps forward 60 (gap : labels 02:20–03:20 never occur).	Fold, gap, and ambiguous labels

Territory	Clock behavior (display vs reference R)	Purpose
Spire	Event-driven: a tick advances only when an event occurs. No duration between events.	Order without duration
Relay	Observer station, identity clock. Signal delay Harbor → Relay = 7 min.	Observation vs occurrence

Events (ground truth)

ID	Territory	True R	Recorded label	Event
E1	Harbor	10	09:10	gate opened
E2	Harbor	40	09:50	traveler entered
E3	Ridge	50	00:50	bell rung
E6	Ridge	80	01:20	lamp lit
E7	Ridge	95	00:35	door barred
E4	Ridge	110	00:50	bell rung
E5	Ridge	170	01:50	vault sealed
S1	Spire	15	tick 1	spire event
S2	Spire	33	tick 2	spire event
S3	Spire	34	tick 3	spire event
S4	Spire	120	tick 4	spire event

Declared causal edges: E1 → E2, E1 → S2, S2 → S3, E2 → S4, E3 → E6, E6 → E7, E4 → E5. **Declared change ledger:** C1 Harbor rate ×2 @R30; C2 Harbor ruleset v4 announced @R90; C3 Ridge fold −60 @R90; C4 Ridge gap +60 @R200 (**undeclared:** no authority record, found only by discrepancy scan); C5 Ridge branch B1 forks @R45.

Note on Ridge labels. Every Ridge label from 00:30 to 01:29 maps to two instants. Ridge events E3, E6, E7, and E4 are therefore all ambiguous when recorded by label alone. This is deliberate.

8. Results

All figures below are read from the output of `mw1_reference_model.py` (deterministic; exact rational arithmetic).

8.1 Regression: reproducing the packet's own numbers

Before running new tests, the model was checked against fixtures other contributors published.

Fixture (source)	Published	MW-1 result	Match
ChatGPT environment A (1×): local elapsed in one reference hour	60 min	60 min	Yes
ChatGPT B (10× acceleration)	10 h (600 min)	600 min	Yes
ChatGPT C (0.25× rate)	15 min	15 min	Yes

Fixture (source)	Published	MW-1 result	Match
ChatGPT D (paused 12:15–12:45)	30 min	30 min	Yes
ChatGPT E (event-driven)	38 event ticks	38 ticks	Yes (by construction)
Perplexity Harbor: displayed minutes in the block	90	90	Yes
Perplexity Harbor: traveler entry label / local minutes spent	09:20 / 70	09:20 / 70	Yes
Perplexity Harbor: 10:00 appointment at reference minute / block-end label	45 / 10:30	45 / 10:30	Yes

The published numbers reproduce exactly, so the clock machinery is trustworthy enough to run new tests on. (M)

8.2 Battery: lab model against a minimal baseline

Fifteen tests, grouped A (label semantics), B (ledger), C (order and knowledge), D (adjudication), E (light and cone), H (tokens). Each test has an oracle taken from MW-1's ground truth. Three outcomes: **PASS**; **GAP** (honestly INDETERMINATE where the fixture has a determinate answer); **WRONG** (a confident wrong answer, or destroyed information). The **lab model** has all seven primitives and a strict reader. The **baseline B0** is a deliberately minimal comparator: one unqualified timestamp per event, overwrite on correction, all labels assumed to share one master clock, and a default-assuming reader. It is not a claim about any real system. Its purpose is to check that the battery has teeth.

Test	What is checked	Lab	B0
A1	Fold: label 00:50 on Ridge -> candidate instants {R50, R110}	PASS	WRONG
A2	Gap: label 02:50 (170) on Ridge -> no instant; flagged unmappable	PASS	WRONG
A3	Label inversion: E6 (label 01:20) vs E7 (label 00:35) -> E6 precedes E7	PASS	WRONG
A4	Same label, distinct events: E3 and E4 (both 00:50) -> collision flagged, not merged	PASS	WRONG
B1	Correction of E1 (R10 -> R8): as-of R50 = 10, as-of R100 = 8, 2 rows kept	PASS	WRONG
B2	Correction of E1 lists dependents {E2, S2, S3, S4}	PASS	WRONG
B3	Remap Harbor v3 -> v4: E2 keeps original 09:50, adds 10:50	PASS	WRONG
C1	Cross-territory order, no shared clock: E1 (Harbor) vs S2 (Spire) -> E1 first (declared edge)	PASS	WRONG
C2	Branch isolation: could E9 (branch B1) have influenced E6 (branch B0)? -> no	PASS	WRONG
C3	Occurrence vs observation: E2 occurs R40, Relay observes R47	PASS	WRONG
D1	Treaty dispute under four dispute rules -> per-rule verdicts, dependence visible	PASS	WRONG
E1	Propagation delay Harbor->Relay stays 7 under display operations	PASS	WRONG
E2	Intervention at E1: dependency cone = {E2, S2, S3, S4}	PASS	WRONG

Test	What is checked	Lab	B0
H1	Audit token, named territories: bounded completeness, undeclared C4 flagged	PASS	WRONG
H2	Token overreach ("own the history", "modify") -> refused	PASS	WRONG

Result (M): the lab model passes 15 of 15. The baseline is wrong on 15 of 15. That confirms only that the battery discriminates between a record-keeping model with the primitives and one without. It does not show that the lab model is better than any real system.

Ablation: what protects what

To trace tests to primitives, I removed one primitive at a time from the lab model (strict reader kept) and recorded which tests stopped passing.

Primitive removed	Tests that stop passing	Failure kind
Clock identity	A1 (GAP), A2 (GAP), A4 (GAP), E1 (GAP)	GAP
Ruleset version	A1 (GAP), A2 (GAP), A4 (GAP), B3 (GAP), D1 (GAP)	GAP
Causal edges	A3 (GAP), B2 (GAP), C1 (GAP), C2 (GAP), E2 (GAP)	GAP
Branch id	C2 (GAP)	GAP
Append-only ledger	B1 (WRONG), B3 (WRONG), D1 (WRONG)	WRONG
Observation records	C3 (WRONG), E1 (GAP)	WRONG + GAP
Authority / rights model	H1 (WRONG), H2 (WRONG)	WRONG

Finding (M). Removing clock identity, ruleset version, causal edges, or branch id degrades answers to **GAP**: the reader says "cannot tell". Removing the append-only ledger, observation records, or the authority model produces **WRONG**: the reader answers confidently and incorrectly, because the information was destroyed rather than merely absent. Missing metadata is recoverable. Overwritten history is not.

Each primitive is caught by at least one test, and no test is an orphan. This is the traceability that neither of the other two frameworks supplies.

8.3 Drift study: what survives a hand-off

The 11 MW-1 claims were passed through four progressively lossy forms and then read by both readers. Three question classes were asked: INSTANT (11 questions), MEMBERSHIP in reference interval [30, 100) (11), and ORDER-REF over all 55 pairs, for 77 questions per level.

- **L0**: full record, with occurrence reference.
- **L1**: occurrence reference dropped; clock, ruleset, label, tick, and causal edges kept.
- **L2**: L1 plus ruleset version, branch, and uncertainty dropped.
- **L3**: L2 plus clock identity and causal edges dropped: a bare label or tick.

Cells are correct / indeterminate / wrong.

Level	Reader	INSTANT (11)	MEMBERSHIP (11)	ORDER (55)	Correct	Indet.	Wrong
L0	Strict	11 / 0 / 0	11 / 0 / 0	55 / 0 / 0	77	0	0
L0	Default-assuming	11 / 0 / 0	11 / 0 / 0	55 / 0 / 0	77	0	0

Level	Reader	INSTANT (11)	MEMBERSHIP (11)	ORDER (55)	Correct	Indet.	Wrong
L1	Strict	3 / 8 / 0	4 / 7 / 0	27 / 28 / 0	34	43	0
L1	Default-assuming	3 / 0 / 8	7 / 0 / 4	32 / 0 / 23	42	0	35
L2	Strict	0 / 11 / 0	0 / 11 / 0	15 / 40 / 0	15	62	0
L2	Default-assuming	3 / 0 / 8	7 / 0 / 4	32 / 0 / 23	42	0	35
L3	Strict	0 / 11 / 0	0 / 11 / 0	0 / 55 / 0	0	77	0
L3	Default-assuming	2 / 0 / 9	7 / 0 / 4	32 / 0 / 23	41	0	36

Results (M). Over all four levels the strict reader gave **0 wrong answers** in 308 questions. The default-assuming reader gave **106**. The strict reader's loss shows as INDETERMINATE growing from 0 at L0 to every INSTANT and MEMBERSHIP question by L3. The default-assuming reader's loss is silent.

Luck is not correctness. At L1 the default-assuming reader gets 3 of 11 INSTANT questions right, but for reasons that do not generalize: Ridge labels E3 and E6 happen to fall on the first candidate, and E5 is unambiguous. The other 8 are wrong. On the four Ridge fold events it is right only when the first candidate happens to be true.

H1 in one example (M). Ridge event E7 (label 00:35) recorded at L1 is INDETERMINATE for INSTANT, because its candidates are R35 and R95. It is answerable for MEMBERSHIP in [30, 100): both candidates fall inside the interval, so the answer is "inside" either way. The same claim is insufficient for one question and sufficient for another.

8.4 Non-commutation of temporal changes

Grok noted that offset, rate, topology, authority, and ontology changes "do not commute". In MW-1's toy algebra I tested four operations on an identity clock, applied in both orders: Offset +60 scheduled at R100; Rate $\times 2$ at R30; Pause of 15 at R50; and Offset +60 scheduled at **local** 02:00 (label 120), i.e. triggered when the local clock reaches a label.

Pair (A $\times$ B)	Commute?	Witness (display at R, A then B vs B then A)
OFF+60@R100 $\times$ RATE $\times 2$ @R30	Yes	—
OFF+60@R100 $\times$ PAUSE15@R50	No	R100: 85 vs 145
OFF+60@R100 $\times$ OFF+60@L120	No	R120: 180 vs 240
RATE $\times 2$ @R30 $\times$ PAUSE15@R50	Yes	—
RATE $\times 2$ @R30 $\times$ OFF+60@L120	No	R75: 180 vs 120
PAUSE15@R50 $\times$ OFF+60@L120	Yes	—

Result (M). Three of six pairs commute. The pattern matters more than the count:

- An offset scheduled in reference time does not commute with a pause. The pause shifts the reference time at which later breakpoints land.

- The **same** offset scheduled in local time does commute with the pause, because it moves with the local clock. It no longer commutes with a rate change, because the rate change alters when the local clock reaches the label.
- An offset scheduled at a label the local clock has already skipped never fires (a change scheduled inside a gap). The order (offset then local trigger) shows this.

This is H4: commutation is a property of the pair **and** the coordinate the change is scheduled in. In a Time War, "effective at 02:00 local" and "effective at reference minute 120" are different weapons. (M; the algebra is my definition, not a discovered property of time.)

8.5 Adjudication: same evidence, four dispute rules

A treaty allows entry "before 10:00 Harbor time". E2 (traveler entered) occurred at R40. The treaty was signed at R5 under v3. Ruleset v4 was announced at R90. The dispute is adjudicated at R200.

Governing rule	Reading of E2	Verdict
rule-at-event (v3 in force at R40)	09:50	Complies
rule-at-agreement (v3 at R5)	09:50	Complies
rule-at-adjudication (v4 at R200)	10:50	Violates
reference-fixed (deadline = R45)	R40 vs R45	Complies

Result (M). Three rules say the entry complies; one (rule at adjudication) says it violates. Nobody disputes that E2 occurred, or when. The dispute is over which ruleset governs. A ledger that returns one verdict conceals this (baseline B0 returned "violates", the answer under the current label). In MW-1 the war is fought entirely on the Settlement ledger. (H5)

8.6 Cone, token, and light

Dependency cone

Intervention at	Cone (records affected)	Cone size	Untouched	Universal recompute
E1	E2, S2, S3, S4	4	6	10
E3	E6, E7	2	8	10

An intervention at E1 propagates to 4 of the other 10 records; at E3, to 2 of 10. Universal recompute touches all 10. (M) The MW-1 graph is sparse by construction, so this shows the mechanism, not a general ratio. (H6)

Light and observation

Relay observes E2 at R47 for an occurrence at R40. The propagation delay of 7 stays exactly 7 under every display operation applied to Harbor or Relay: offsets, rate changes, and pauses. Delay is computed in reference coordinates, so a civil-clock change cannot alter it. A model that computes the delay by subtracting stored labels gets −543 (test E1, B0). (M, consistent with E: a light-year is a distance, and civil time zones relabel observation without changing the distance.)

The four ways an origin-of-light change can be meant, adopted from Perplexity, need four different ledger writes: a change to the unit's definition (IL), a change to how signals propagate after a boundary event (a new propagation ruleset, IL, plus consequences inside the cone), a change to surviving records (new CL rows), and a change to history itself (S: a new branch or a declared

mutation, whichever decision 9 chooses). The test fails if any two become indistinguishable in the ledger.

Token scope

Request	Outcome
Audit right, territories {Harbor, Ridge}	READ ONLY. 5 changes returned. Completeness bounded to Harbor + Ridge. Change C4 flagged as having no authority record.
Audit right, no territory, fail-closed policy	REFUSED (no territory set). Nothing returned.
Audit right, no territory, annotate policy	READ ONLY. 5 changes returned. Flagged: territory set unspecified; Spire not audited; no completeness claim.
"Own the history"	REFUSED. The asset class is not defined in the instrument.
Modify right, {Harbor}	REFUSED. Grants are fail-closed.

Result (M). Removing the authority model produces **WRONG** on both token tests (Section 8.2): it returns an implicitly complete history, and grants what was asked. The annotate policy for reads costs nothing in rights and returns more than fail-closed, at the risk of being mistaken for a complete answer. The completeness flag is what prevents that. (H7)

8.7 Hypothesis verdicts

ID	Verdict inside MW-1	Evidence
H1	Supported	E7 at L1: INSTANT indeterminate, MEMBERSHIP determinate. MEMBERSHIP determinate for 4 claims vs INSTANT for 3 at L1.
H2	Supported	Strict reader: 0 wrong in 308. Default-assuming reader: 106 wrong.
H3	Supported	Ablating append-only breaks B1, B3, D1, all with WRONG answers.
H4	Supported	The same offset commutes with a pause when scheduled locally and does not when scheduled in reference time.
H5	Supported	3 rules comply, 1 violates. B0 returned a single verdict.
H6	Supported (weakly)	Cone 4 of 10 and 2 of 10, on a sparse hand-built graph.
H7	Supported	Grants refused in every case; reads returned 5 changes with flags; ablating authority produced WRONG .

8.8 Run log

Two corrections were made before the final run, and are recorded here instead of being applied silently.

- **Oracle error.** In an early run test H1 failed because its oracle expected 4 changes; the in-scope corpus (Harbor + Ridge) holds 5. The oracle was wrong, not the model. Corrected to 5.
- **Baseline outcomes were first hard-coded.** The initial comparator table asserted the baseline's failures instead of computing them. That would have been an assertion, not a measurement. The baseline was reimplemented as a switchable model and its outcomes are now computed by the same oracles as the lab model's.

9. Packet drift ledger

The multi-AI packet is itself a specimen for the drift taxonomy. The table records observed movement in what each claim supports. Several rows are legitimate design moves. The ledger records movement, not fault. I assess against the frameworks' own criteria, not against real-world science.

Source	Move	Drift	Repair
Gemini	Validates events by stamping ticks since an epoch; a "meta-ledger" holds the absolute tick.	DR4: an order/duration count carries the role of a presentation label. Assumes one shared meta-clock.	Ticks are per-clock. Add clock identity (test A1, ablation of clock).
Gemini	The higher-permission or last edit "becomes canonical truth".	DR5: settlement presented as evidence.	Settlement row that cites claims (LR3).
Copilot	Causality check $t(E2) > t(E1)$, "in the relevant timeline".	DR3: the scope is in the prose but not in the invariant. Labels can invert under a fold (A3).	State the invariant on order coordinates, with timeline and frame as parameters.
Copilot	A master timeline and per-universe conversion f_i are defined up front.	DR9: silent answer to decision 1 (privileged clock?).	Mark as an open decision. Keep mappings local until a settlement act.
ChatGPT	The rewritten prompt adds an event-validation framework and a required-output list beyond the dictation.	DR3 at prompt level: the execution answered a larger question than was asked.	Diff each rewrite against the original dictation before execution.
Claude (earlier)	Three stamps: happened, observed, recorded.	DR1: omits ruleset version, frame, uncertainty, and dependency cone (self-audit; Grok's critique accepted).	This document.
Perplexity	Invariant 3: mappings cannot place B before A.	DR3: true of order coordinates, not display labels.	Scope to order coordinates (Section 3.1).
Grok	<code>unit_kind</code> is a single field on the marker.	DR4 risk: one label can serve different kinds of question.	Attach kind to the question class (Section 5.3).

10. Time War implications

10.1 Attack surface by ledger

Attack	Ledger	What an honest ledger still holds	Residual (undetectable case)
Stamp without binding	CL	The claim with no binder; classified weak, not deleted.	A well-forged binder.
Ruleset omission (tick 500, version 7 vs 8)	IL	Both rows with distinct ruleset ids.	Records that never carried ruleset ids.

Attack	Ledger	What an honest ledger still holds	Residual (undetected case)
Fold collision, gap erasure	CL / IL	Candidate sets: fold gives 2 instants, gap gives 0 (tests A1, A2, A4).	Events recorded only by label in a territory with no clock history.
Rate arbitrage (treaty in local labels)	SL	Per-rule verdicts (test D1).	A settlement that names no rule.
Overwrite "correction"	CL	Prior belief via as-of query (test B1). Destroyed only if LR1 is broken.	Complete destruction of the ledger.
Settlement laundering	SL → CL	The SL row has no write path to CL (LR3).	A reader who does not distinguish SL from CL.
Token overclaim	SL	The grant is refused (test H2).	A recognized authority that grants it.
Carrier-chain flattening	all	The original record, if the chain kept it. The drift meter reports what the derived record supports.	A derived record that is the only surviving copy.

10.2 Every verb writes one ledger

Operation	Writes to	Cannot write
Relabel (display, calendar)	IL	Event, claim
Remap (ruleset version)	IL	Event, claim; earlier IL rows
Correct knowledge	CL (new claim) + IL (link)	The original claim
Reobserve	CL (observation row)	Occurrence
Reclock (rate, offset, pause)	IL (new clock ruleset)	Payload of events
Resettle / re-adjudicate	SL	CL, IL
Branch	CL (new branch id)	Rows of the parent branch
Alter history (fictional)	CL (new event rows on a new branch, or a declared mutation)	The record that it happened

10.3 A scenario in this framework

The Harbor treaty dispute (8.5) is a complete small Time War. **Parties:** the Harbor authority holds the ruleset change (IL); the traveler's guild holds the original gate log (CL); Relay holds an independent observation (CL); an adjudicator holds the Settlement ledger. **Contested event:** E2, which nobody disputes. **Evidence:** the gate log at 09:50 under v3; the reference log; the v4 decree announced at R90 with retroactive reinterpretation; Relay's observation at R47. **Possible decisions:** any of four rules, giving three "complies" verdicts and one "violates". **Unresolved:** which rule the treaty means, which is a decision the architect makes (Section 11, decision 6), not something the ledger can decide.

11. Architect decision register

These are yours. The lab runs its battery under each option; it does not choose. Silent defaults become fake physics (DR9).

#	Decision	Options	What the lab runs under each
1	Is there a privileged reference clock in-world?	None; optional declared; mandatory	Re-run all questions with and without a candidate reference. Count INDETERMINATE answers.
2	Which layers can an authority change?	Display; rate; experienced duration; causal graph; settlement	For each layer, confirm the ledger write it produces (10.2).
3	Do changes apply prospectively, in place retroactively, or by branch?	Three	Re-run B1, B3, C2 for each.
4	What can a token grant, and who recognizes it?	Read/audit; custody; traverse; govern; modify	Re-run H1, H2 per grant class. Elevated grants must fail closed.
5	What evidence survives a rewind or branch, and who may inspect it?	All; branch-local; sealed	Re-run B1, C2 with each survival rule.
6	Which dispute rule governs?	At event; at agreement; at adjudication; reference-fixed	Run D1: 3–1 split in MW-1 shows the choice matters.
7	Are cross-territory effects immediate, signal-limited, or special?	Three	Re-run E1 and observation tests.
8	Do bodies and processes follow local rate, reference rate, or a third rule?	Three	Add an ageing claim class; test DISPLAY vs elapsed.
9	Is the past mutable, forkable, or presentation-only editable?	Three	Re-run E2 (cone) and C2 for each.
10	May residuals remain permanently?	Yes; no; with expiry	Count residuals across reconciliation runs.
11	Reader policy for reads vs grants	Fail-closed everywhere; annotate reads and fail closed on grants (proposed)	H1, H2, and token study for each.
12	Who may write the Settlement ledger, over which territory?	Single; overlapping; disputed	Run D1 with two competing authorities to produce two SL rows for one event.

Provisional draft recommendation (not canon): claims immutable; interpretations and settlements append-only; no in-world master clock until a story or protocol needs a settlement act; grants fail closed, reads annotate; fictional rewind or branch permitted only as a labeled ledger write.

12. Limits and unresolved questions

What MW-1 cannot show

- **Circularity risk.** I designed the battery and the model. Tests derived from the principles they test can pass because I wrote them to. Mitigation: the multi-contributor challenge in Phase 3, where other contributors run the same fixtures. Until then, results carry label M only.

- **Toy algebra.** The commutation results depend on how I defined the operations. They show that non-commutation is coordinate-dependent, not how time behaves.
- **Small n.** Eleven events, three territories. The cone result depends on a sparse hand-built graph.
- **Hidden ground truth.** MW-1 scores answers against a reference R that no in-world reader has. If canon has no such ground truth, INDETERMINATE answers cannot be checked.
- **No adversary.** Section 10.1 describes attacks; MW-1 does not yet run an adversarial reader or forger.

Open paradoxes (retained)

- Two frames can disagree on the order of distant events. A war that demands one order demands a convention.
- An event-driven world has no duration between events, so a contract written in hours has no native meaning there.
- An honest audit of "every change" needs a closed universe of territories and records. An open world cannot sell that completeness.
- If a presentation can be rewritten faster than inhabitants can sense it, "what they thought the time was" becomes another contested archive.
- Buying a mapping does not buy the processes that already ran under the old mapping.

13. Next phases

1. **Phase 1 (done).** MW-1 v0.1, battery, ablation, drift study, regression against published fixtures.
2. **Phase 2.** Expand MW-1: more territories, a second branch, event-driven and relativistic-style observers, longer causal graph. Run the battery under each architect decision (Section 11).
3. **Phase 3: AI-as-reader drift test.** Give each contributor the same 11 claim records at levels L0–L3 and the same 77 questions, with the instruction to answer or say INDETERMINATE. Score against the strict reader's oracle: correct, indeterminate, wrong. This measures directly what the packet review could only describe. It also tests whether contributors answer where the record is insufficient. I can supply the fixture file and scoring script.
4. **Phase 4.** Adversarial runs (Section 10.1) against the reference model.
5. **Phase 5.** Promotion decisions: which objects move toward canon, recorded with dissent, after Phases 2–4.

Appendix A. Contribution and provenance

Idea	Origin	Use here
Temporal stack of layers (physical, civil, environment, simulation, logical, observation, record, valid, knowledge, narrative)	ChatGPT	Checklist of what a claim can carry (5.3)
Three operations: change representation / correct knowledge / alter history	ChatGPT	Basis of the ledger-per-verb table (10.2)
Dependency cone instead of universal rewrite	ChatGPT (also Perplexity)	H6, test E2
Temporal jurisdiction	ChatGPT	Decision 12
Environments A–E for the one-hour block	ChatGPT	Regression fixtures (8.1)
Light-cone framing of a light-year war	Gemini	Context for 8.6
Per-universe conversion function, fixed points	Copilot	Context for decision 1 and 9
Mutable vs branching past; attack-vector list	Claude (earlier response)	Decision 9; Section 10.1
Labels E / D / S; decision list; bounded-corpus completeness; query-centered reconciliation	Perplexity TITF	Sections 1, 8.6, 11
Unit kinds; operations table; "unknown is data"; presentation/environment split; conversion-and-dispute surface	Grok TLF	Sections 4, 10.2; question classes
Question-relative sufficiency (candidate-set rule)	This document	5.3, H1
Drift taxonomy, strict and default-assuming readers, hand-off chain	This document	5.4, 5.5, 8.3, 9
Three ledgers and LR1–LR5	This document	5.2
Commutation depends on (operation, scheduling coordinate)	This document (builds on a Grok remark)	8.4, H4
Ablation traceability	This document	8.2
Fail-closed grants, annotated reads	This document	5.3, H7
AI-as-reader drift test	This document	Phase 3

Appendix B. Glossary

Term	Meaning
Claim	An immutable assertion that an event occurred, with its markers, frame, ruleset, and evidence.
Candidate set	All reference instants a claim could denote given the fields it retains.
Fold / gap	A backward or forward clock jump that makes labels occur twice or not at all.
Sufficiency	A claim is sufficient for a question if every candidate reading gives the same answer.
Strict reader	Answers only when all candidates agree; otherwise INDETERMINATE.

Term	Meaning
Default-assuming reader	Fills every gap with a default; never says INDETERMINATE.
Drift	Change in what a claim supports after a transformation.
Cone	The set of records depending on an event through declared causal edges.
Ledger separation	Claims, interpretations, and settlements live in separate append-only ledgers.

Appendix C. Reproducing the results

Run `python3 mw1_reference_model.py`. It is deterministic and prints every table used in Section 8, and writes `results.json`. Regression fixtures (8.1) run first. Any change to a clock definition, edge list, or oracle should be logged as a new version of the model, with the earlier version kept.

Verbatim source note. This document paraphrases the two reviewed frameworks. For any disagreement between my summary and their text, their text governs.